Machine Learning Analysis of English Premier League Matches

Allen, Zachary

Allen.642@wright.edu

McCabe, Tyler

Mccabe.37@wright.edu

Pradhan, Sulav

Pradhan.49@wright.edu

Abstract- This report explores historical trends in the English Premier League (EPL) by applying a multi-faceted machine learning approach to a comprehensive dataset of match results. Using statistical analysis, we performed dimensionality reduction via Principal Component Analysis (PCA) to identify influential match features and employ K-means clustering to categorize match performance profiles. Furthermore, the study incorporates predictive analysis by implementing logistic regression, neural networks, and gradient-boosted tree models (XGBoost) to forecast match outcomes. By comparing unsupervised clustering with supervised classification across multiple model families, this research demonstrates how latent data structures, rolling-form features, and non-linear patterns can be leveraged to understand and predict the competitive dynamics of Association Football.

GitHub Link: https://github.com/zmallen794-source/Premier-League-Predictive-Analysis

I. Introduction

The English Premier League is one of the most unpredictable sports environments in the world. This study addresses the problem of identifying significant performance trends and predicting future outcomes with a dataset from 1993 to 2022. While traditional statistics focus on simple win/loss ratios, this research seeks to uncover the latent structures of match dynamics and build robust predictive frameworks

We contribute a systemic approach to match analysis by:

- Identifying the statistical features that define Premier League match variance through PCA
- Splitting matches into the following profiles: “Home Team Dominance/Win”, “Away Team Dominance/Win” and “Competitive/Draw”
- Developing predictive models using logistic regression
- Extending the predictive framework with XGBoost ensemble models and rolling-form feature engineering across five experiments to compare model families
- Create a neural network that can challenge the baseline set by XGBoost

II. Methodology

1. Unsupervised Learning

Using PCA, we transform the 16-feature dataset into a 3-dimensional space to capture maximum variance. A K-Means algorithm (with n_init=10 and random_state=42) is then applied to group matches into k=3 clusters based on these components

2. Supervised Learning

- **Logistic Regression**: A probabilistic model to determine linear relationships between match statistics and result probabilities
- **XGBoost Ensemble Models:** Gradient-boosted decision tress applied across with successive experiments applying anti-overfitting techniques and rolling-form feature Engineering
- **Neural Networks**: Utilized to capture complex, non-linear interactions between features and try to match the outcome baseline set by XGBoost model.

3. **Advanced Feature Engineering – Rolling Team Form**

A central innovation in the extended experiments is the computation of rolling averages over the last 5 games for each team. Initial analysis revealed that using match-day statistics to predict that same game's outcome introduces critical "look-ahead bias.". This rolling form captures the pre-game form, a team's recent performance trajectory, which is a well-established predictor in sports analytics.

The data is processed chronologically, computing a rolling average over the previous five matches. for: goals scored/conceded, shots and shots on target, corners, and fouls; producing 12 engineered features per match. Matches where a team has fewer than 5 historical games are filtered out. In later experiments, Points (3/1/0 for win/draw/loss) and Goal Difference were added, extending the feature set to 16 features.

A window of 5 was chosen to balance signal recency with statistical stability. Using team-agnostic statistics rather than team identifiers allows models to generalize across teams rather than memorizing team-specific patterns.

III. Experimental Setup

A. Datasets

The study utilizes the results.csv dataset, containing several thousand Premier League games. Features include full-time and half-time goals (FTHG, FTAG, HTHG, HTAG) offensive volume (Shots, Shots on Target, Corners), and disciplinary metrics (Fouls, Yellow/Red Cards).

After applying rolling-form feature engineering, additional features were introduced to enhance the performance of both the XGBoost and neural network models.

B. Implementation Setup

All features were standardized using StandardScaler prior to dimensionality reduction and model training to ensure that high-magnitude features did not disproportionately bias the PCA or gradient-boosted and neural network models over low-magnitude features.

The data was split chronologically (no random shuffling) to prevent data leakage across time, using 70% for training, 15% for validation, and 15% for a held-out test set for logistic regression, the XGBoost and neural network experiments.

C. Evaluation Metrics

To evaluate all gradient boosted and Neural network models, performance was assessed using Accuracy, Precision, Recall, F1-Score, Macro Average, Weighted Average, Multi-Class Log Loss (mlogloss), and Validation Loss

The combined use of accuracy, macro-averaged F1, and per-class recall is

essential for this problem because the Draw class is both the smallest and the hardest to predict. A model with high accuracy but zero Draw recall (as observed in Experiment 1) is functionally deficient despite appearing statistically acceptable.

IV. Results

A. Unsupervised Learning – K-means and PCA

When reviewing the PCA loading, one can see what is affecting the outcome of games the most.

For PC1, the features that affect the most are Away Shots, Home Shots, and Home Shots on Target. For PC2, the top 3 features are Full-Time Home Goals, Half-Time Home Goals, and Away Shots on Target. For PC3, Half-Time Away Goals, Away Yellow Cards, and Away Foul are their primary features. Just looking at the top 3 in each of the PCs, the data seems to be relatively even in terms of what differentiate between match types (Home win, Away win, or Tie).

	PC1	PC2	PC3
AS	0.407446	0.248312	-0.087826
HS	-0.391553	0.211812	0.180015
AST	0.357985	0.333355	0.086800
HST	-0.352105	0.322742	0.166563
FTAG	0.286111	0.326562	0.347579
HC	-0.280727	0.032000	0.224291
AC	0.272410	0.116392	-0.225621
FTHG	-0.257944	0.418945	-0.027110
HTAG	0.208327	0.296213	0.388335
HTHG	-0.168592	0.387037	-0.072906
HY	0.144194	-0.157795	0.333441
HR	0.121158	-0.013144	0.172287
HF	0.096295	-0.194692	0.286204
AR	-0.095199	-0.039474	0.178023
AY	-0.028351	-0.181946	0.388761
AF	-0.026461	-0.215129	0.374329

Figure 1. The loadings of each PCA

However, since PC1 has a little bit more weight, it would stand to reason that the Away Team's performance affects the results of the match slightly more than the home team.

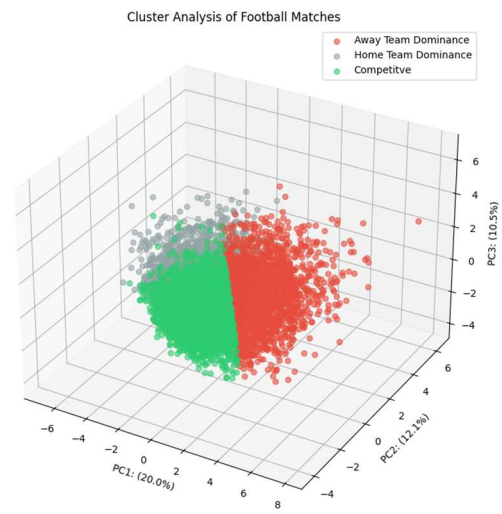

Figure 2. PCA-transformed match data

Figure 2 shows how each match type was clustered. While the location for each cluster can be a bit random, each grouping in this model was able to accurately separate between the three match types.

---Cluster Summary Table (Feature Average) ---

ClusterLabel	ATD	Competitve	HTD
FTHG	1.243044	0.972191	2.640672
FTAG	2.142390	0.691827	0.820336
HTHG	0.616203	0.370034	1.223180
HTAG	0.998773	0.247162	0.383025
HS	10.779051	12.127128	18.463593
AS	15.038052	9.245743	8.142180
HST	4.835925	4.885074	9.337355
AST	7.445990	3.669977	3.616114
HC	4.734861	5.885074	7.798794
AC	6.356792	4.452043	3.632917
HF	11.584288	12.145857	9.998707
AF	11.095336	12.822645	11.251616
HY	1.640753	1.568672	0.947006
AY	1.558101	1.976447	1.595864
HR	0.125614	0.050511	0.015511
AR	0.041735	0.095630	0.130116

Figure 3. The average values for each feature for each cluster.

In the "Away Team Dominance" cluster, all the away team performance parameters are better than the home team performance parameters; the reverse is true for the "Home Team

Dominance" cluster. This holds true all the way down to the home vs Away Red Cards. The Competitive, as expected, has the closest Full-Time Goal differential out of the three clusters. Both values are also below 1, indicating that a lot of these might end with a 0-0 or 1-0 result.

For comparison, the precision, recall, and F1-score were also reviewed from the clusters developed in the unsupervised learning. Since this was unsupervised learning, all the data was considered "training," so it can be assumed that this would be the theoretical max for K-means for this model:

	Precision	Recall	F1-score	Support
Away Win	0.58	0.59	0.58	2417
Tie	0.35	0.59	0.43	2069
Home Win	0.79	0.48	0.6	3803

Table 1. Max Classification Value from K-Means Clustering

B. Predictive Performance – Logistic Regression

For the purposes of predictive modelling, the variables can be divided into factors that can be known pre-match, at half time of the match, or post-match. The pre-match variables include factors such as who is playing, who the referee is, and when the match is taking place; the half time variables entail the Half Time Score for the Home and Away Teams, as well as the Half Time Result, while the post-match variables entail the information concerning fouls, shots taken, and the final scores. The final scores shall be ignored for the purposes of predictions, as they have a 1:1 correlation to the winner when taken together for logistic regression.

Looking at the pre-match statistics, the highest precision achieved is with 3 variables, combining the HomeTeam, the Ref, and the Season, though it only achieves a precision of 49%. The logistic regression appears to heavily favor predicting a home team victory, which tracks with its higher occurrence in the dataset, though it never predicts a tie, resulting in its overall poor performance.

Moving to the stats known at half-time, The HTHG and HTAG are the most important stats, and they achieve a precision of 62%. This comes with a greater propensity to predict away team wins, though it again fails to successfully predict ties.

Of the final statistics, HST, AST, and AC carry the most impact in predicting the winner, reaching a precision of 58%. The model fails to predict ties yet again, and more heavily favors predicting away team wins, resulting in inaccuracies in regard to home team wins.

C. Predictive Performance – XGBoost

Experiment 1 – Baseline XGBoost
(Best params:
Learning rate = 0.01
Max depth = 5
N estimators = 200)

	Precision	Recall	F1	Support
Home Win (0)	0.52	0.89	0.66	564
Away Win (1)	0.47	0.34	0.39	358
Draw (2)	0.20	0.00	0.01	305
Accuracy			0.51	1227
Macro Avg	0.40	0.41	0.35	1227
Weighted Avg	0.43	0.51	0.42	1227

Table 2. Evaluation metrics for baseline XGBoost Model

Misleading higher accuracy 51% achieved by ignoring the draws.

Experiment 2 – Class-Weighted XGBoost
(16 features, balanced sample weights):

Class	Precision	Recall	F1	Support
Home Win (0)	0.59	0.49	0.53	564
Away Win (1)	0.41	0.52	0.46	358
Draw (2)	0.24	0.23	0.23	305
Accuracy			0.43	1,227

Macro Avg	0.41	0.41	0.41	1,227

Table 3. Evaluation metrics for Class weighted XGBoost Model

Balanced performance. Draw recall improved dramatically from 0.00 to 0.23. The model now makes more diverse predictions rather than defaulting to the majority class.

Experiment 3 — Heavily Regularized XGBoost
(Best params:
Column sampling=0.8,
Learning rate=0.01,
Max depth=9,
N estimators=300,
subsample=0.8):

Class	Precision	Recall	F1	Support
Home Win (0)	0.57	0.57	0.57	564
Away Win (1)	0.43	0.51	0.46	358
Draw (2)	0.24	0.19	0.21	305
Accuracy			0.46	1,227
Macro Avg	0.41	0.42	0.42	1,227

Table 4. Evaluation metrics for Regularized and class weighted XGBoost Model

Accuracy improved slightly to 46% compared to Experiment 2, and the macro F1 reached 0.42, the best among the XGBoost experiments when considering all classes equally. There is no clear improvement, suggesting the model reached its performance ceiling.

D. Predictive Performance – Neural Network

Experiment 1 – Shallow Architecture
(Hyperparameters:
Width = 256
Dropout = 0.2
Learning rate = 0.001
Batch size = 32
Epochs = 80)

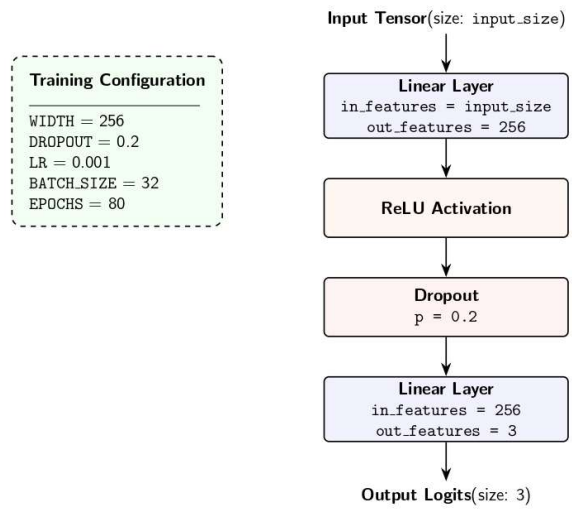

Figure 4. Architectural diagram of the shallow neural network

The shallow wide model suffered from overfitting issues due to the combination of high model capacity (256 hidden units) and the size of dataset samples. Instead of generalizing the model effectively memorized the data. This can be seen in the learning curves, where training loss rapidly decreases but the validation loss spiked.

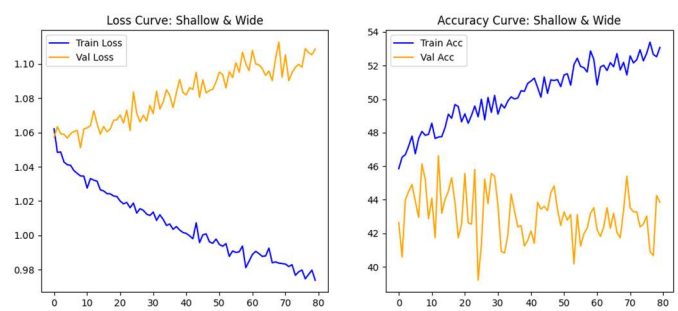

Figure 5. Learning Curves showing the loss and accuracy of shallow model

Experiment 2 – Deep Architecture
(Hyperparameters:
Hidden layer = 64
Dropout = 0.3
Learning rate = 0.0005
Batch size = 64
Weight decay = 1e-4
Epochs = 100)

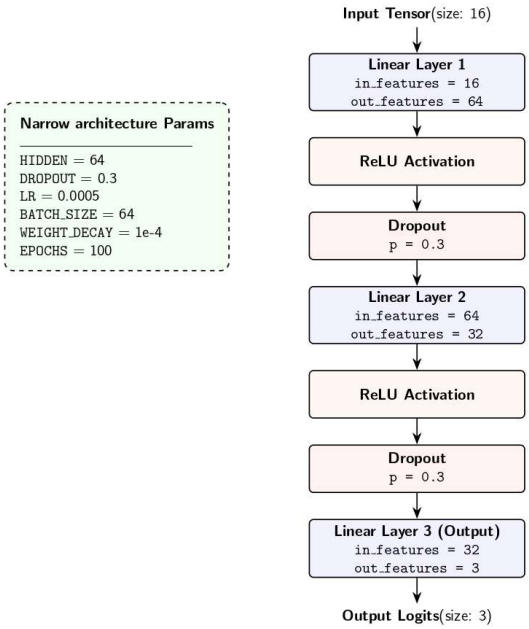

Figure 6. Architectural diagram of the deep neural network

While the gap between training and validation curve was still significant, the validation loss did not spike up like in shallow model. This showed that deep architecture was better in generalizing than shallow architecture. Hence, deep architecture was chosen for further experimentation.

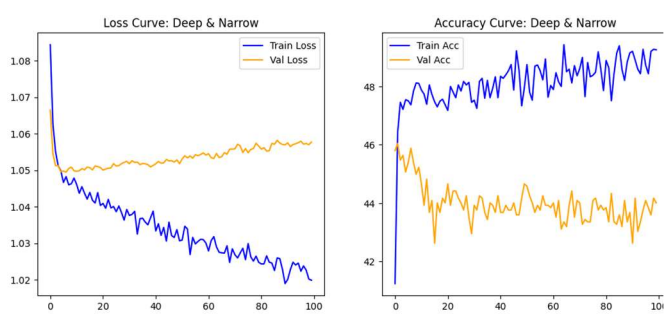

Figure 7. Learning curve showing loss and accuracy of Deep model.

Experiment 3 – Heavily Regularized Neural Network Model
(Hyperparameters:
Hidden layer=64,
Dropout=0.5,
Learning rate=0.001,
Batch size=128,
Weight decay=1e-3,
Epochs =100)

The implementation of aggressive regularization techniques, specifically a dropout rate of 0.5, weight decay of 1e-3, and batch normalization, successfully mitigated overfitting. This decreased the gap between the training and validation losses (less than 0.02) maintained throughout training. However, validation accuracy plateaued, oscillating between 43.68% and 45.31% without establishing a clear upward trajectory. This stagnation strongly suggests that the model has reached its performance ceiling given the current feature set. The final validation accuracy of approximately 44% aligns closely with the results achieved by the class-weighted XGBoost.

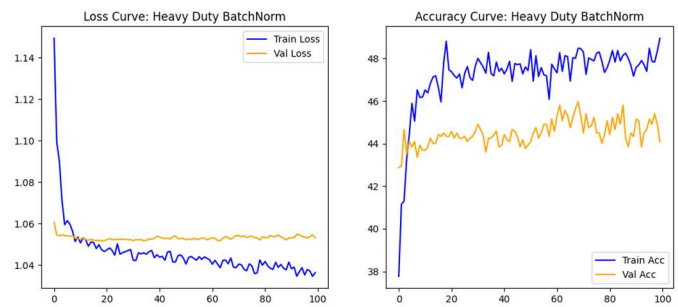

Figure 8. Learning curve showing loss and accuracy of heavily regularized Deep model.

Experiment 3 – Heavily Regularized Neural Network Model
(Hyperparameters:
Same as Heavy Regularization with ReduceLROnPlateau scheduler (patience=5, factor=0.5))

Analysis of the training process shows that the learning rate scheduler rapidly decayed the learning rate from 0.001 to near zero by epoch 50. This confirms that the observed loss plateau is a genuine convergence of the model, rather than an artifact of an excessively high learning rate causing instability. Validation accuracy peaked early

at 45.6% (epoch 10) and ultimately stabilized at approximately 45%. The automated checkpointing mechanism correctly identified epoch 29 as the optimal model state. Because this overall performance is virtually identical to the Heavy Regularization variant, it confirms that optimization parameters like learning rate are no longer the bottleneck; the model is strictly limited by the predictive power of the current feature set.

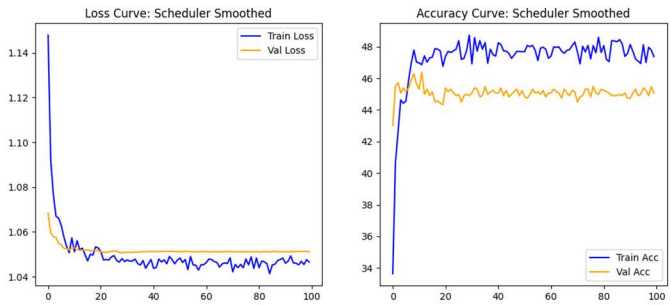

Figure 9. Learning curve of heavy regularized deep model with dynamic learning rate scheduler

Class	Precision	Recall	F1	Support
Home Win (0)	0.59	0.54	0.56	564
Away Win (1)	0.41	0.54	0.46	358
Draw (2)	0.24	0.19	0.21	305
Accuracy			0.45	1,227
Macro Avg	0.41	0.42	0.42	1,227
Weighted Avg	0.45	0.45	0.45	1227

Table 5. Evaluation metrics for Regularized Neural Network Model

Accuracy was comparable to the XGBoost model. The model also predicted draws, preventing lazy prediction of 0 Recall in draws.

E. Comparison between best XGBoost model and best Neural Network model

The XGBoost model holds a marginal advantage with better recall and demonstrates superior interpretability through its detailed classification report. However, from a practical standpoint, the two approaches perform comparably.

When considering the baseline XGBoost (Experiment 1), its 51% accuracy appears substantially higher. However, this figure is misleading because it reflects the model's tendency to overwhelmingly predict the majority class (Home Win), effectively ignoring Draws entirely (F1 = 0.01). The class-weighted and regularized XGBoost models sacrifice raw accuracy for genuinely balanced class predictions, which is more meaningful for real-world applications.

The NN models achieved ~45% accuracy with minimal overfitting (train-validation loss gap < 0.02), suggesting they reached a stable generalization performance.

Class	Precision	Recall	F1-Score	Support
Home Win (0)	0.59	0.58	0.58	564
Away Win (1)	0.45	0.55	0.49	359
Draw (2)	0.32	0.25	0.28	305
Accuracy			0.49	1228
Macro Avg	0.45	0.46	0.45	1228
Weighted Avg	0.48	0.49	0.48	1228

Figure 6. Evaluation metrics of best obtained XGBoost model

Class	Precision	Recall	F1-Score	Support
Home Win (0)	0.59	0.58	0.58	564
Away Win (1)	0.45	0.55	0.49	359
Draw (2)	0.32	0.25	0.28	305
Accuracy			0.49	1228
Macro Avg	0.45	0.46	0.45	1228
Weighted Avg	0.48	0.49	0.48	1228

Figure 4. Evaluation metrics of best obtained Neural Network model

V. Discussion

The results confirm that scoring metrics and shot accuracy are more significant indicators of match trends than disciplinary actions. The

higher loading value in PCA for Away Team Performance indicates that the away team dictates the result of the match more than the home team does. It also appears to be Competitive/Ties when there is the lowest amount of scoring occurring.

The predictive results for logistic regression do make logical sense, as it is hard to predict a match prior to kick off without knowing more factors, i.e. injuries or outs. The mid-match hinging on the home and away scores is also a result of a lack of factors, as knowing the number of penalties or shots taken by the half could improve the predictive power. The final prediction being driven by the shots taken by both teams also tracks, as a team taking more shots is more likely to score, and thus more likely to win. Disregarding the final scores is necessary, as that would be more of an if-then statement than a prediction.

For the XGBoost and Neural network experiments, both model plateaus at approximately the same accuracy (~45%). This strongly suggests that our rolling average feature worked well and represents a natural ceiling of predictable information.

The XGBoost experiments reveal a clear trade-off: raw accuracy (Experiment 1, 51%) versus balanced class performance (Experiment 3, 46% with macro F1 = 0.42, heavy penalization for missing draws). The neural network experiments, with uniform heavy regularization, bypass this trade-off but land in the same performance range. The near-zero train-validation loss gap in all NN experiments confirms proper generalization; the bottleneck is the feature set, not the architecture.

XGBoost and Neural Networks achieved comparable performance on this structured tabular dataset, confirming the well-established empirical finding that gradient-boosted trees are competitive with or superior to neural networks on moderate-sized tabular data. XGBoost holds advantages in speed, interpretability, while neural networks offer greater flexibility for larger, more complex feature spaces.

One thing that is apparent across all the models was predicting ties. This emphasized by the precision and recall values from the XGBoost, but was true for PCA and Logistic Regression as well.

VI. Conclusion

This study demonstrates combining unsupervised learning with supervised predictive modeling provides a structured and interpretable framework for analyzing English Premier League match dynamics. PCA and K-means clustering successfully revealed underlying performance profiles, distinguishing between home wins, away wins, and draws, while highlighting the importance of shot-based and scoring-related features in defining match outcomes.

On the predictive side, logistic regression established a useful baseline but struggled with class imbalance and nonlinearity, particularly failing to capture draws. More Advanced models- XGBoost and neural networks- achieved substantially improved and more balanced performance, especially after incorporating class weighting and rolling-form feature engineering. However, both approaches ultimately converged to a similar performance ceiling of approximately 0.42-.45 macro F1-scores.

This convergence is a key finding: it suggests that model architecture and optimization are not primary limiting factors. Instead, the predictive ceiling is driven by the inherent uncertainty of football outcomes and the limitations of the available feature set. Even with carefully engineered rolling averages, the model lack access to critical contextual variables such as tactical decisions and in-

game dynamics, which are likely necessary to achieve further performance gains.

Additionally, the persistent difficulty in predicting draws across all models highlights a fundamental challenge in sports analytics, where minority classes are both underrepresented and structurally harder to model.

Overall, this work reinforces the effectiveness of gradient-boosted trees for structured tabular data while showing that neural networks can achieve comparable performance when properly regularized. More importantly, it emphasizes that meaningful improvements in predictive accuracy will likely require richer, more context-aware data rather than increasingly complex models. Future work should explore the integration of player-level data, betting odds, or real-time match features to push beyond the observed performance ceiling and better capture the complexity of match outcomes.

AUTHOR CONTRIBUTIONS

Zachary Allen: *Abstract, Introduction, Unsupervised Methodolgy, Unsupervised Learning Results and Discussion, Conclusion*

Tyler McCabe: *Logistic Regression Methodology, Results, and Discussion*

Sulav Pradhan: *Neural Network and Boosted Tree Models. Introduction, Methodology, Experimentation, Results, and Discussion related to these models*